IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

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Executive Summary

- Summary of methodologies
  - Collected and validated SpaceX Falcon 9 launch data.
  - Cleaned and prepared dataset, performed EDA, SQL Quries and Visualizations
  - Build interactive insights , developed and tuned model.
  - Evaluated the model.
- Summary of all results
  - Landing success rate increased overtime
  - Payload mass and launch site strongly influence outcomes
  - Identified key drivers of landing success for decision support
  - Classification model achieve reliable prediction performance

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Introduction

- Project background and context
  - SpaceX pioneered reusable rocket technology to reduce launch costs
  - Falcon 9 first stage landing is critical for mission economics
  - Historical launch pads enable data driven performance analysis
  - Data science helps to uncover patterns behind landing success
- Problems you want to find answers
  - Which factors most impact first stage landing success?
  - Can landing outcome be predicted before launch?
  - How do payload , orbit and launch site affect results?
  - Which ML model predict outcome best ?
  - How can insight support operational and cost decisions?

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Section 1

Methodology

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Methodology

Executive Summary

- Data collection methodology:
  - Using data set from provided source , fetched data using pandas , verified data integrity and consisitency
- Perform data wrangling
  - Converted null values , added Categorical values, Normalized and cleaned payload and launch variable.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Applied hyperparameter tuning, cross validating , compared performance across parameter

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Data Collection

- Describe how data sets were collected.
  - Source: History SpaceX Falcon 9 launch dataset, imported and loaded data using pandas
  - Verified structure, schema, and completeness
  - Stored dataset for preprocessing and analysis
- You need to present your data collection process use key phrases and flowcharts
  - Data Source
  - Data Import
  - Data Validation
  - Data Storage
  - Output

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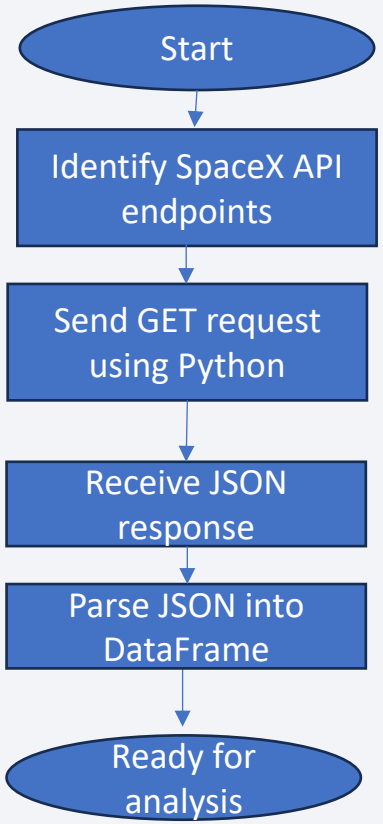
Data Collection – SpaceX API

Key phrases and flowcharts

- Accessed SpaceX REST API for launches and rockets
- Used Python requests library to fetch JSON data
- Parsed and transformed JSON into structured DataFrames
- Cleaned and validated data for analysis

External Reference:
Peer-review the complete SpaceX API notebook at:

- https://github.com/sreeramvijayan/Data-Science/blob/main/jupyter-labs-spacex-data-collection-api%20(2).ipynb



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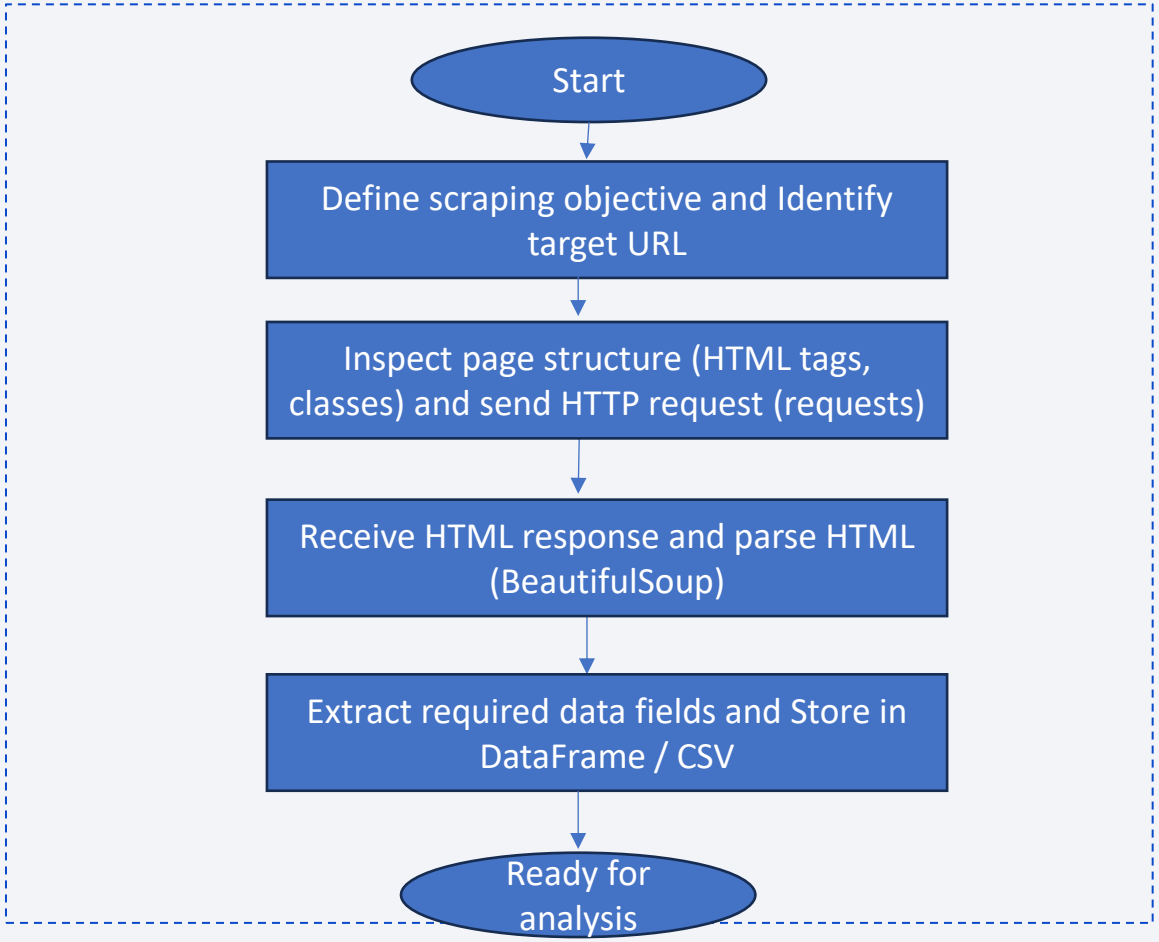
Data Collection - Scraping

Key phrases and flowcharts

- Target website identification
- Inspect HTML structure
- Send HTTP request
- Extract and store required elements in structured format (DataFrame/CSV)
- Data cleaning & validation
- Ready for analysis / visualization

External Reference:
Peer-review the complete SpaceX API notebook at:

- https://github.com/sreeramvijayan/Data-Science/blob/main/jupyter-labs-webscraping%20(1).ipynb



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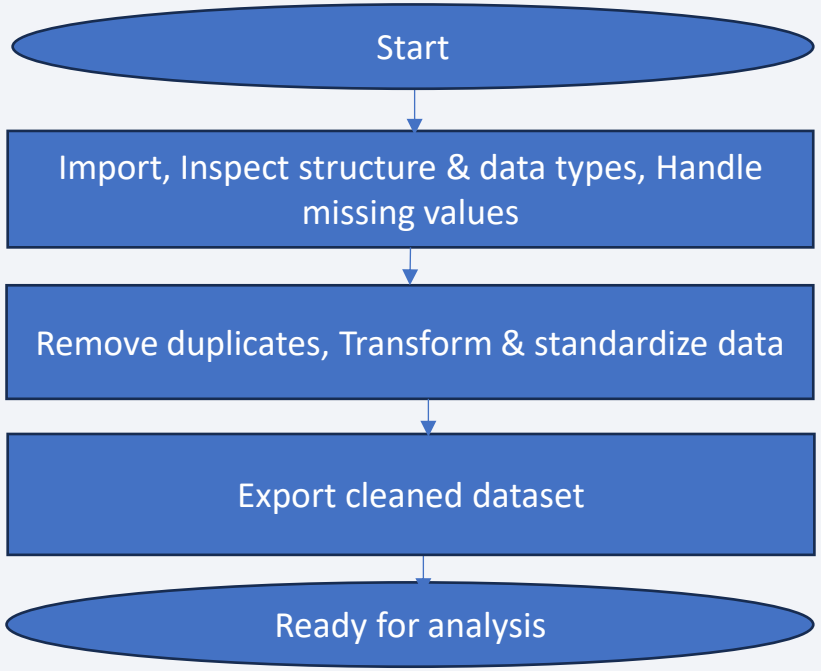
Data Wrangling

- Describe how data were processed
  - Imported raw data and converted to data frame. Checked structure and types and cleaned missing/ null values. Removed duplicated standardized format. Validated and saved clean dataset.
- You need to present your data wrangling process using key phrases and flowcharts
  - Data Ingestion and inspection
  - Handling missing values Removes duplicates
  - Data type conversion and Feature selection
  - Data transformation, Outlier detection and Export cleaned dataset



- External Reference:
  Peer-review the complete SpaceX API notebook at:
  - https://github.com/sreeramvijayan/Data-Science/blob/main/labs-jupyter-spacex-Data%20wrangling%20(1).ipynb

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EDA with Data Visualization

- Summarize what charts were plotted and why you used those charts
  - **Flight Number vs Launch Site : Type:** Strip plot (catplot) : **Reason:** To see which launch sites were used over the sequence of flights and identify patterns in site usage.
  - **Payload Mass vs Launch Site** : **Type:** Strip plot (catplot) : **Reason:** To check if launch sites differ in the payload sizes they handle.
  - **Success Rate by Orbit Type (Bar chart):** compares average success across orbits
  - **Flight Number vs Orbit Type (Strip plot):** visualizes orbit distribution over flights
  - **Payload Mass vs Orbit Type (Strip plot):** explores payload variation by orbit
  - **Yearly Launch Success Trend (Line chart):** shows average success trend over time.

- External Reference:
  Peer-review the complete SpaceX API notebook at:

https://github.com/sreeramvijayan/Data-Science/blob/main/edadataviz.ipynb

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EDA with SQL

- Using bullet point format, summarize the SQL queries you performed
  - **Task 1:** Retrieved unique launch site names using SELECT DISTINCT "Launch_Site".
  - **Task 2:** Displayed 5 records with launch sites starting with 'CCA' using WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5.
  - **Task 3:** Calculated total payload mass for NASA (CRS) missions using SUM("PAYLOAD_MASS__KG_").
  - **Task 4:** Computed average payload mass for booster version F9 v1.1 using AVG("PAYLOAD_MASS__KG_").
  - **Task 5:** Found the date of the first successful ground pad landing using MIN(Date) with WHERE Landing_Outcome LIKE '%ground pad%'.
  - **Task 6:** Listed boosters with successful drone ship landings and payload mass between 4000–6000 kg using WHERE Landing_Outcome='Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000.
  - **Task 7:** Counted total successful and failed missions using COUNT(*) grouped by Mission_Outcome.
  - **Task 8:** Identified booster versions that carried the maximum payload mass using a subquery with MAX(PAYLOAD_MASS__KG_).

External Reference:
Peer-review the complete SpaceX API notebook at:

- https://github.com/sreeramvijayan/Data-Science/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

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Build an Interactive Map with Folium

- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
- Explain why you added those objects
  - Created a Folium base map centered near NASA Johnson Space Center for geographic context.
  - Added Folium Circles at each launch site with popups showing site names to highlight locations.
  - Placed Markers with DivIcon labels on circles to clearly identify launch sites.
  - Used MarkerCluster to group launches, color-coded green for success and red for failure.
  - Added MousePosition plugin to show dynamic latitude and longitude for analysis.
  - Combined elements provide an interactive map for comparing sites, success patterns, and spatial relationships.

External Reference:
Peer-review the complete SpaceX API notebook at:

https://github.com/sreeramvijayan/Data-Science/blob/main/lab_jupyter_launch_site_location.ipynb

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Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
- Explain why you added those plots and interactions
  - **Launch Site Pie Chart** – shows percentage of successful launches per site; updates dynamically when a site or “All Sites” is selected.
  - **Payload Range Progress Bar** – indicates the current payload range in kilograms to filter data interactively.
  - **Payload vs Launch Outcome Scatter Plot** – plots payload mass (kg) on y-axis against launch outcome class on x-axis for selected site(s), with a legend for clarity.
  - **Interactive Controls** – dropdown for launch site selection and slider for payload range, enabling real-time updates of charts for better analysis.
  - **Purpose** – provides an interactive way to explore success rates, payload impact, and site comparisons visually and intuitively.

External Reference:
Peer-review the complete SpaceX API notebook at:

https://github.com/sreeramvijayan/Data-science-Needed-Assignment/blob/main/lab%20dash%20board.png

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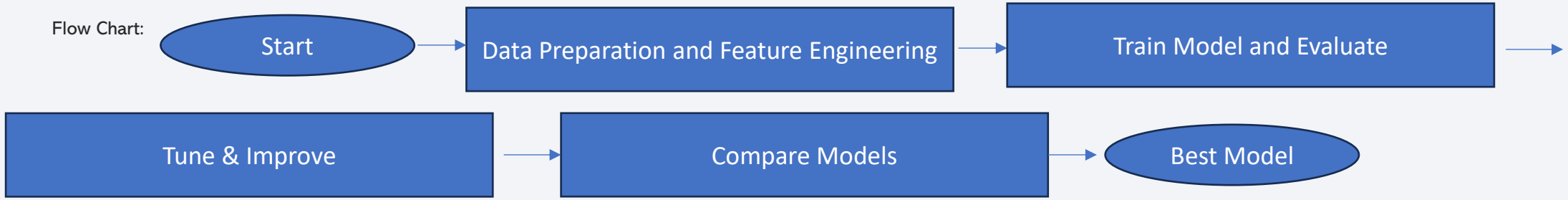
Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model
- You need present your model development process using key phrases and flowchart
  - **Built the model** by defining the objective, preprocessing data, engineering features, and training a baseline on train-test split.
  - **Evaluated and improved performance** using metrics (Accuracy, Precision, Recall, F1, ROC-AUC), cross-validation, feature selection, and hyperparameter tuning.
  - **Selected the best model** through comparison, validation results, and finalizing the pipeline with the highest generalization performance.

Flow Chart:



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- https://github.com/sreeramvijayan/Data-Science/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20(2).ipynb

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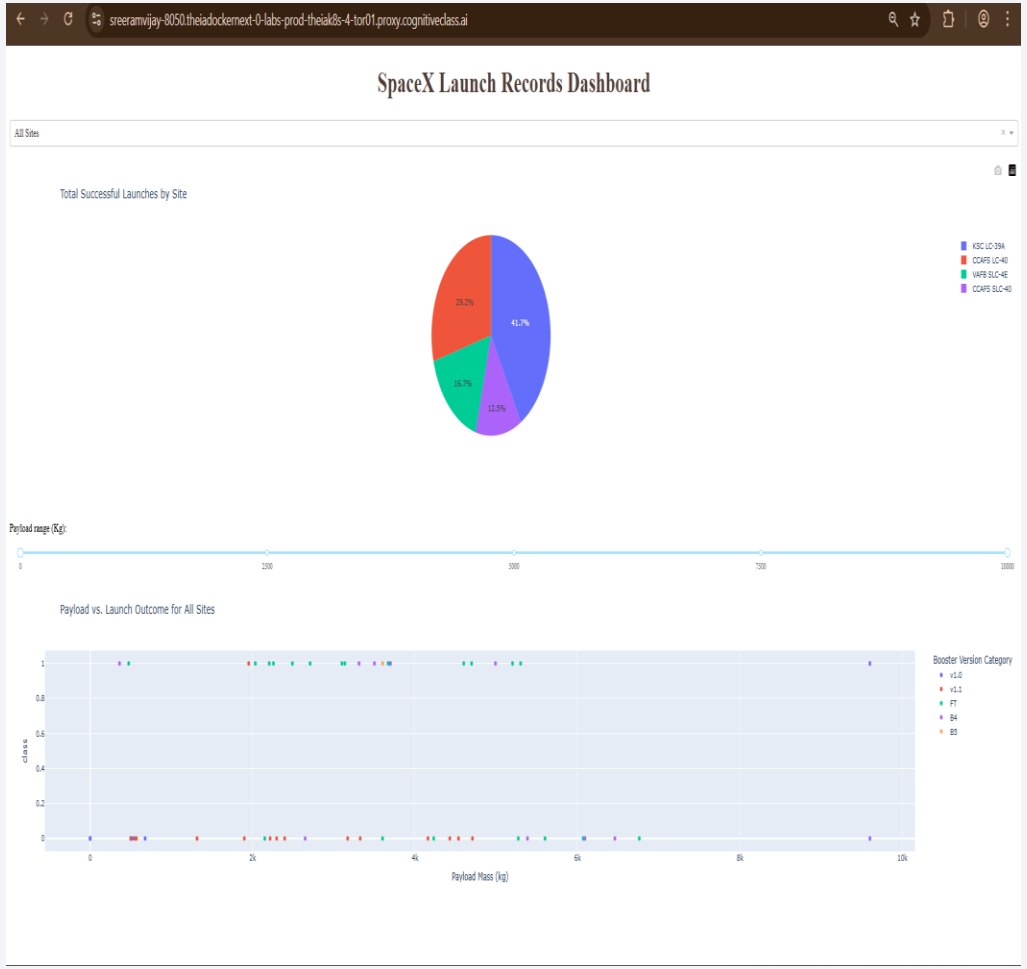
Results

- Exploratory data analysis results
  - Identified key feature distribution and trends
  - Detected correlations between variables
  - Found missing values and outliers
  - Highlighted important predictors
- Interactive analytics demo in screenshots
- Predictive analysis results
  - Best model achieved highest accuracy
  - Matrix map created with the created predicted table with true tables
  - With KNeighbourClassifier as estimator , in GridSearchCV function ,
  - Got the the accuracy using best_score_, with best score .
  - Strong ROC-AUC and balanced F!-score, predicted on unseen data.



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Section 2

Insights drawn from EDA

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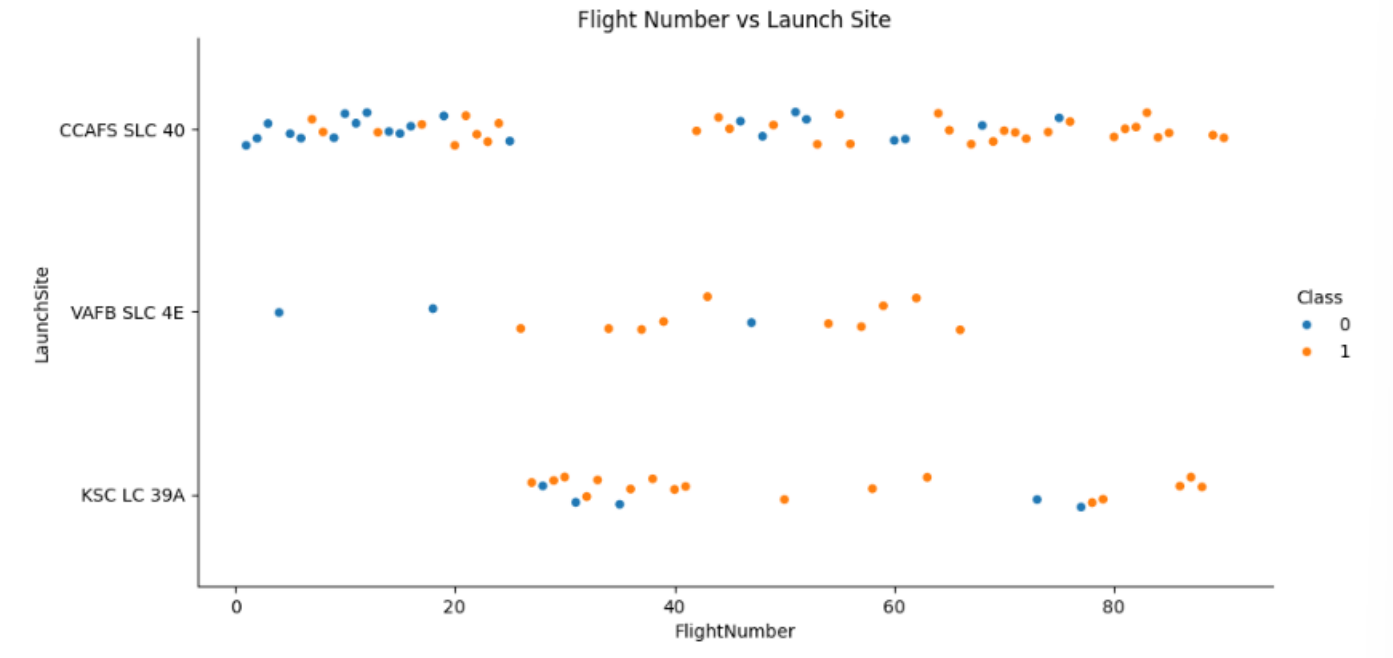
Flight Number vs. Launch Site

Launch success increases with flight number

CCAFS SLC-40 had the highest launch frequency, KSC LC-39A shows mostly successful recent launches, VAB SLC-4E has fewer launches.



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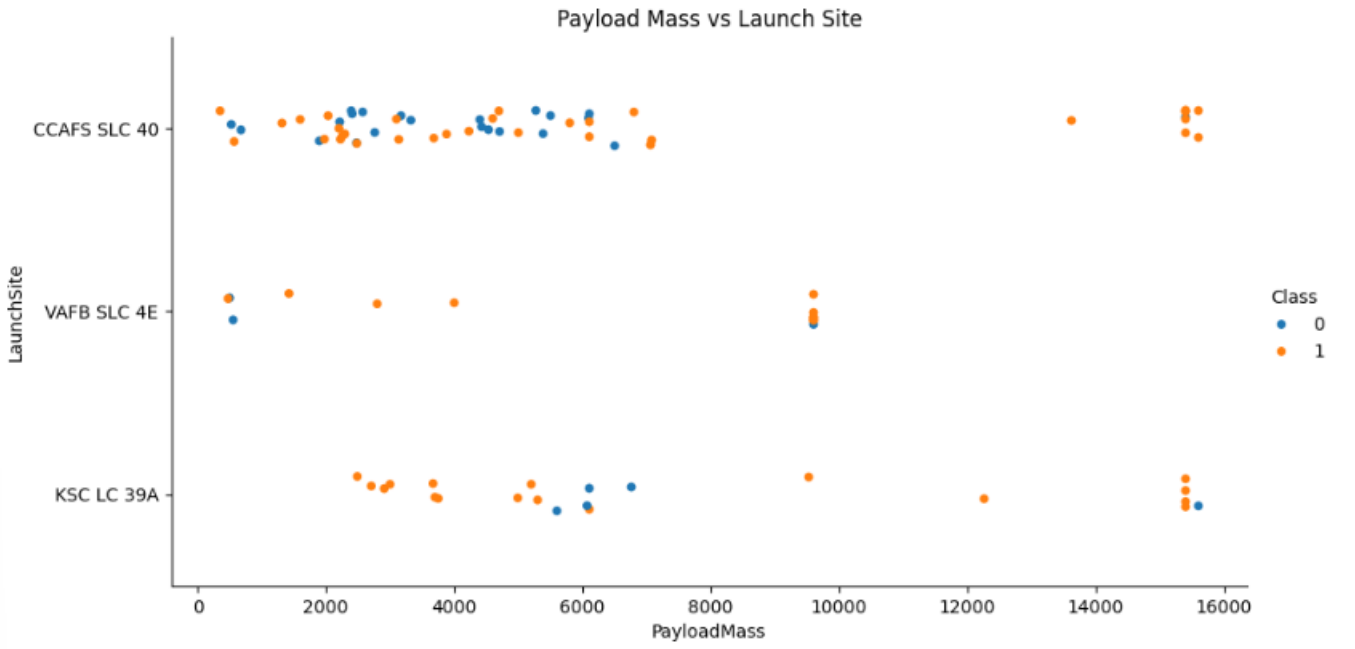
Payload vs. Launch Site

Higher payloads generally show more successful launches

CCAFS S:C-40 handles a wide range of payloads, KSC LC-39A is used for heavier payload mission, VAFB SLC-$E has fewer and mostly mid range payloads.



Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

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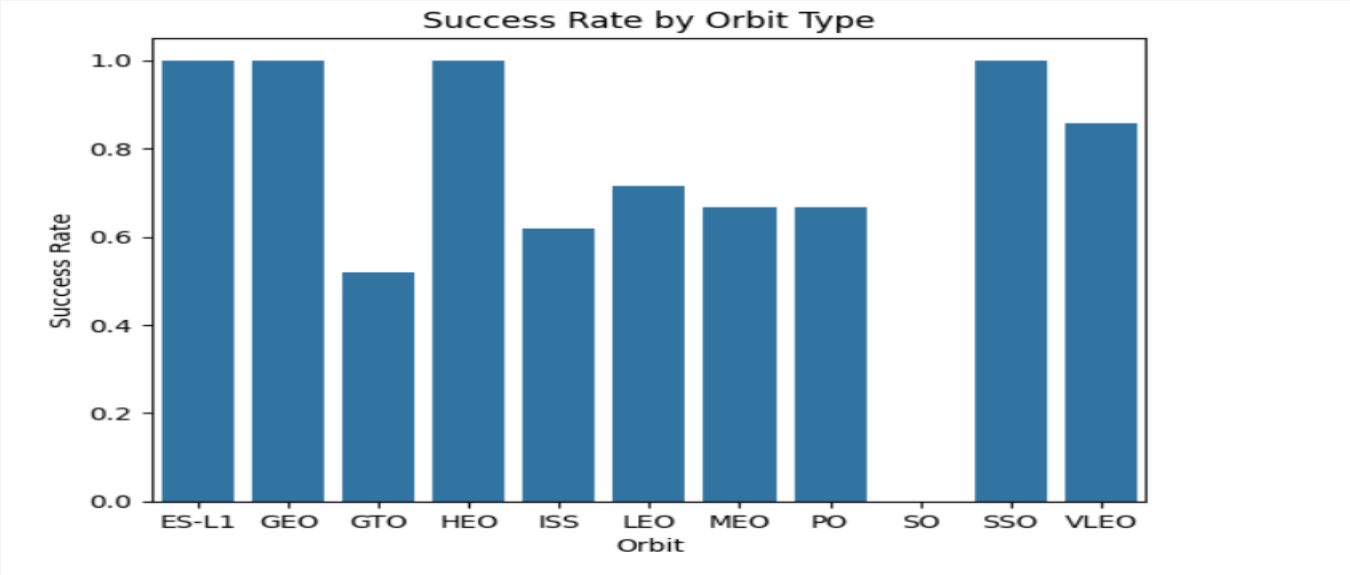
Success Rate vs. Orbit Type

- Highest success rates (≈100%):ES-L1, GEO, HEO, and SSO show near-perfect success → highly reliable mission profiles.
- Moderate success rates:LEO, MEO, and PO fall in the mid range (~65–72%) → consistent but with some variability.
- Lowest success rate: GTO has the loweest success (~50%) → more complex missions with higher risk.
- Most orbit types demonstrate strong reliability, but mission complexity impacts success probability.



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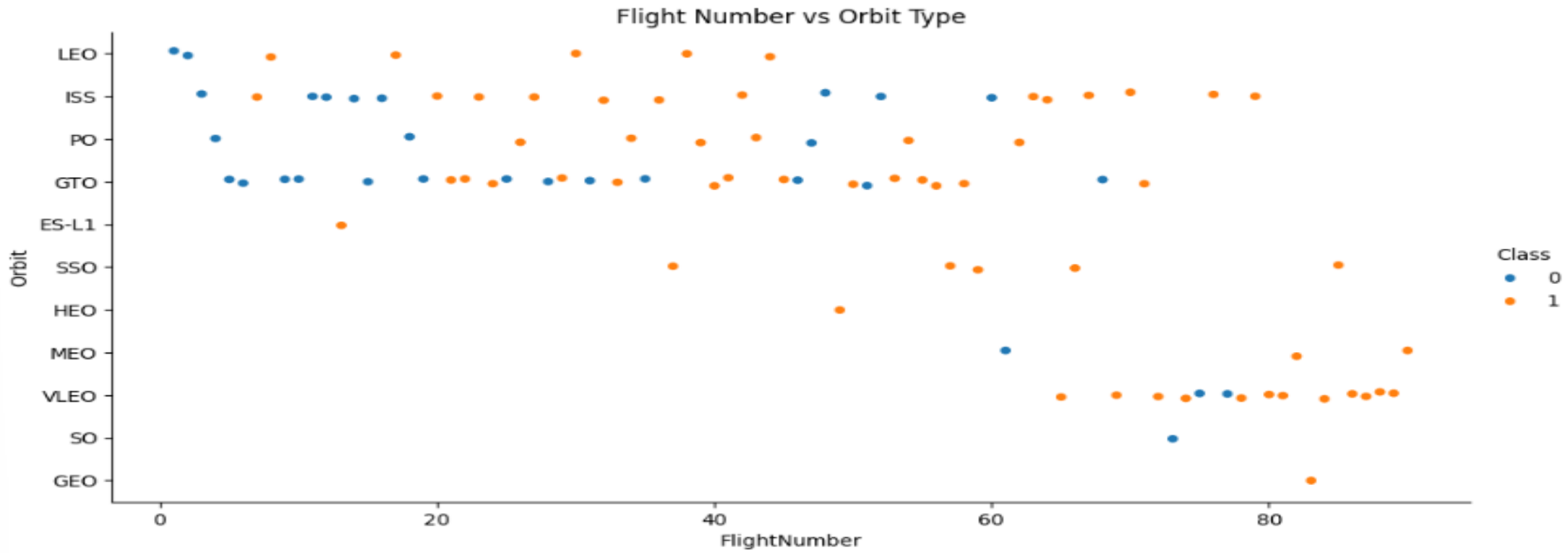
Flight Number vs. Orbit Type



You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

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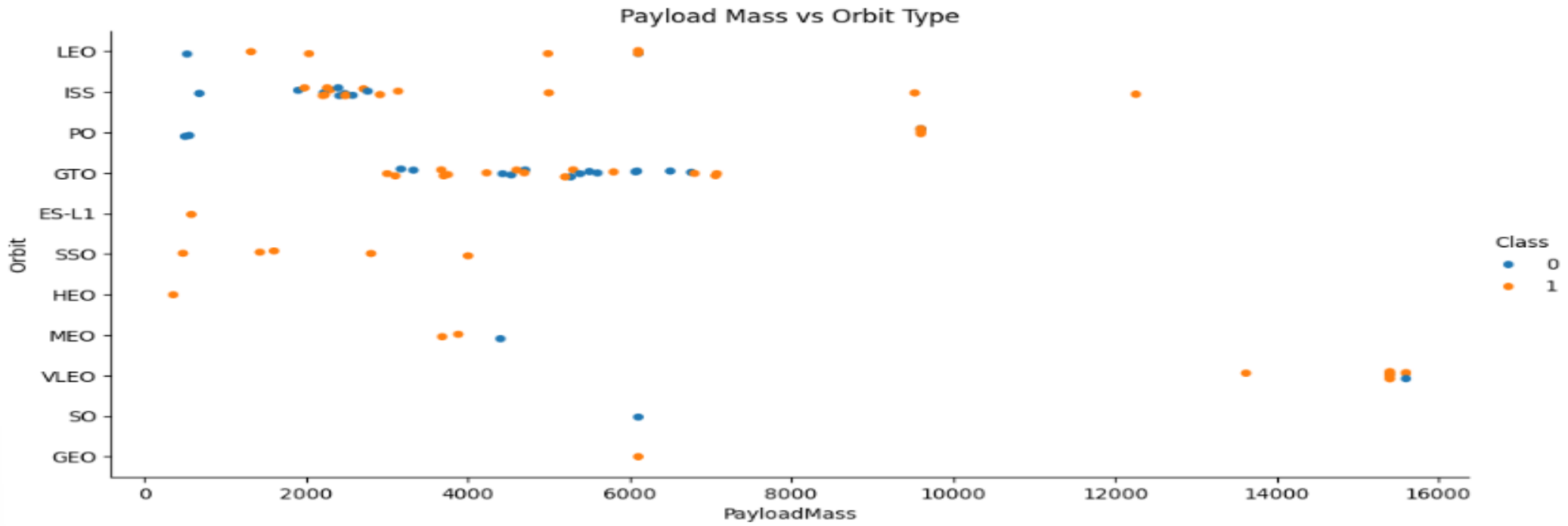
Payload vs. Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.

However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

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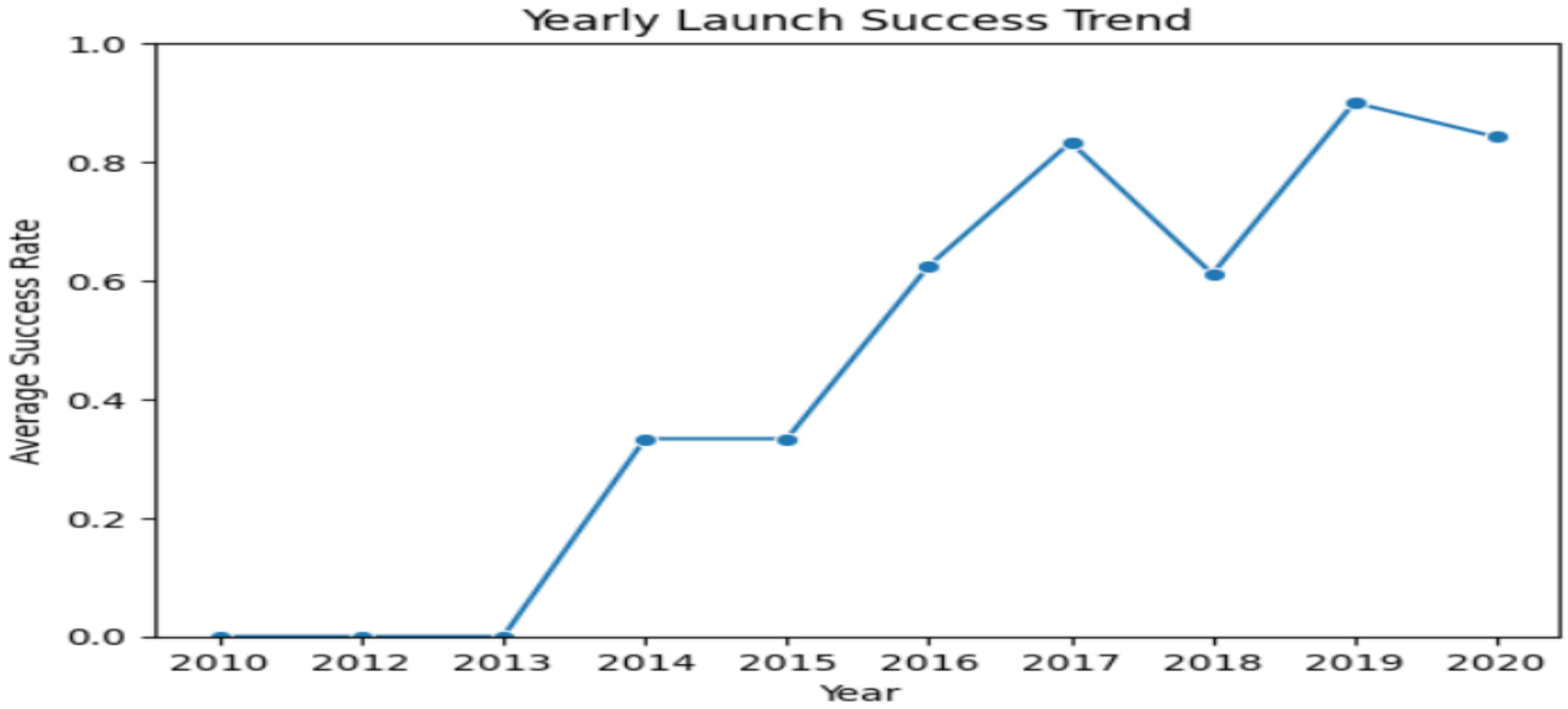
Launch Success Yearly Trend



you can observe that the sucess rate since 2013 kept increasing till 2020

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All Launch Site Names

- **Launch_Site**

  CCAFS LC-40

  VAFB SLC-4E

  KSC LC-39A

  CCAFS SLC-40

- Query to get the unique launch site in the space mission is as follows:
  - SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE

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Launch Site Names Begin with 'CCA'

SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_ _KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese		LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)		

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Total Payload Mass

- Total payload mass carried by boosters launched by NASA (CRS) is 48213.
- The query is : SELECT SUM("PAYLOAD_MASS__KG_") AS TotalPayloadMass FROM SPACEXTABLE WHERE "Customer" LIKE '%NASA (CRS)%';
- F9 v1.0 B0006 and F9 v1.0 B0007 are the booster_version of NASA (CRS).

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Average Payload Mass by F9 v1.1

- Average payload mass carried by booster version F9 v1.1 is 2534.6666666666665
- This is the query to get the result SELECT AVG("PAYLOAD_MASS__KG_") AS AveragePayloadMass FROM SPACEXTABLE WHERE "Booster_Version" LIKE 'F9 v1.1%';

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First Successful Ground Landing Date

- The first successful landing outcome on ground pad is on 2015-12-22
- The query to get the result is SELECT MIN(Date) AS FirstSuccessfulGroundLanding FROM SPACEXTABLE WHERE Landing_Outcome LIKE '%ground pad%';

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Successful Drone Ship Landing with Payload between 4000 and 6000

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

The query for this result is : SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;

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Total Number of Successful and Failure Mission Outcomes

The query for the result is : SELECT Mission_Outcome, COUNT(*) AS Total FROM SPACEXTABLE GROUP BY Mission_Outcome;

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

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Boosters Carried Maximum Payload

- The query is : SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

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2015 Launch Records

- The Query is :SELECT substr(Date,6,2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date,0,5)='2015' AND Landing_Outcome LIKE 'Failure (drone ship)%';

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

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Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The query is : SELECT Landing_Outcome, COUNT(*) AS Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Count DESC;

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

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Section 3

Launch Sites Proximities Analysis

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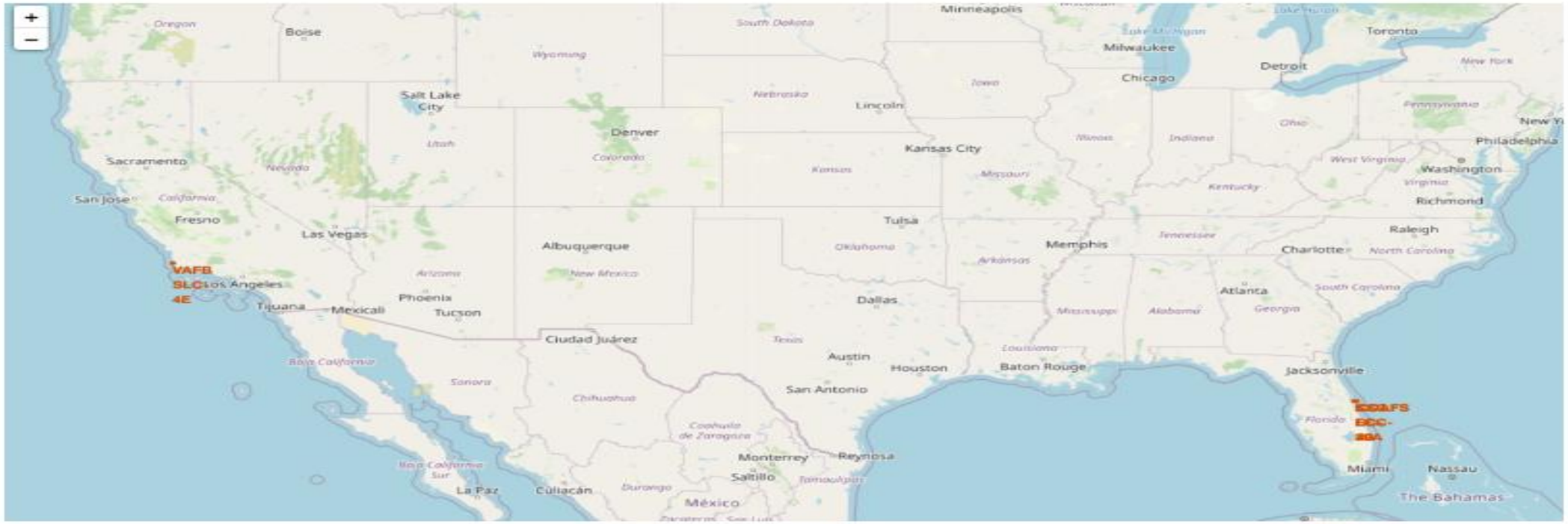
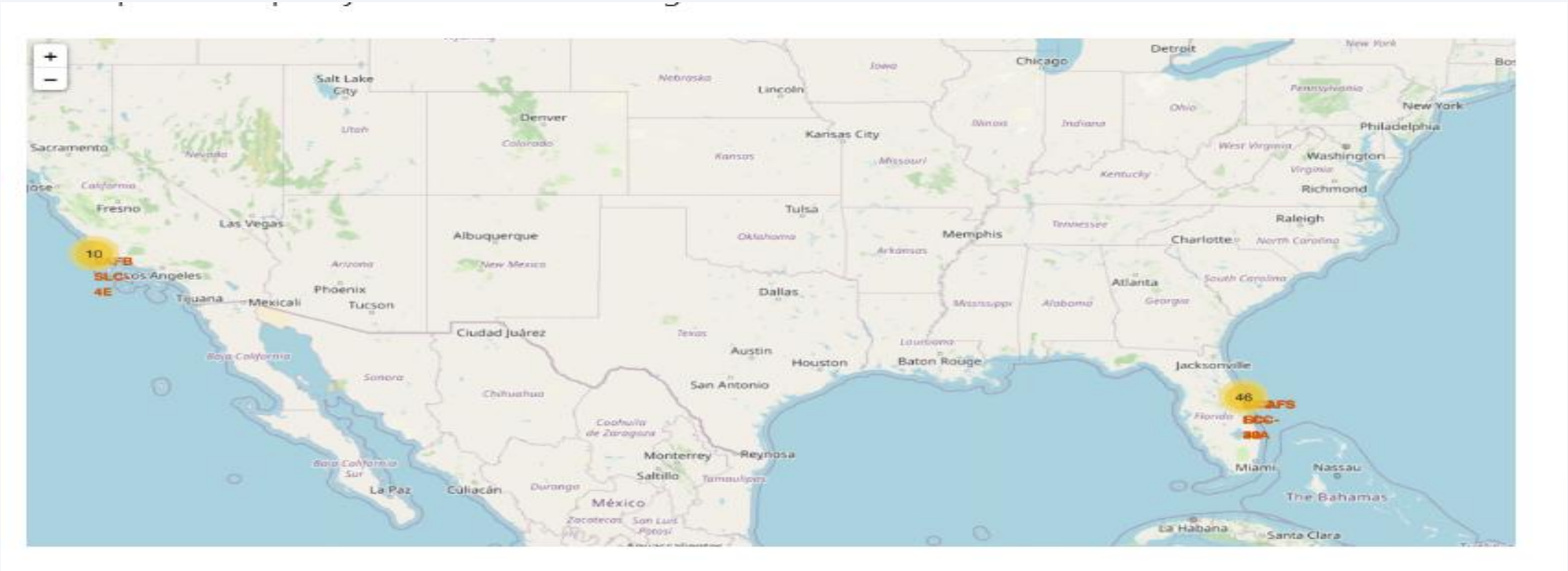
Folium Map with Launch Sites

- The Lauch site in proximity to Equator line and coast.



Folium Map with Launch site Count

- This map indicates the total number of launch sites an success/Failure rates in each launch site



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Folium Map with Launch site Success/Failure Count

- This map indicates the total number of launch sites an success/Falure rates in each launch site

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Section 4

Build a Dashboard with Plotly Dash

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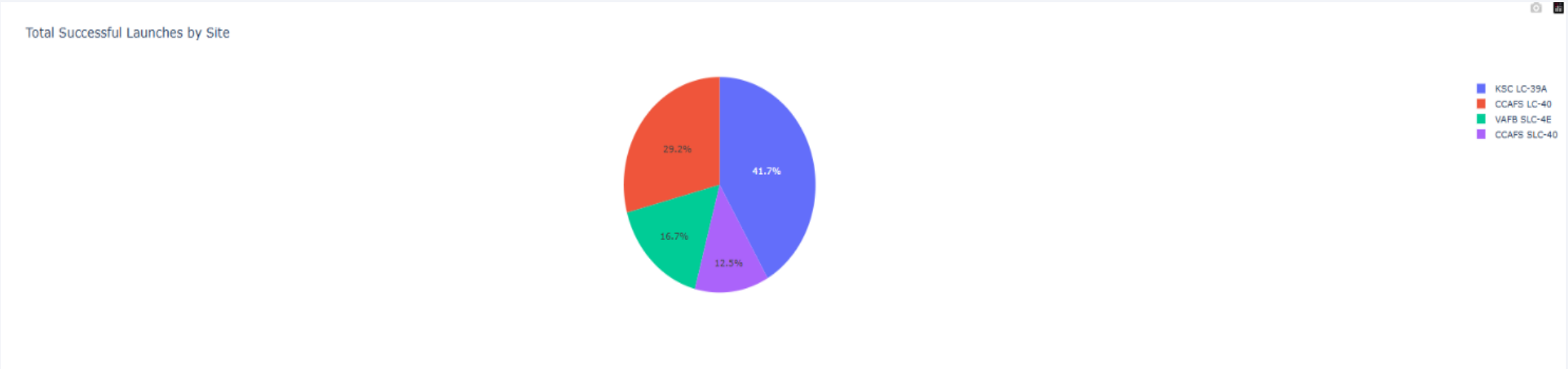
Pie chart with Total Successful Launches by Site

- Pie chart explain the total successful launches by site . From the chart we can say that KSC LC-39A has the highest successful launch in all sites



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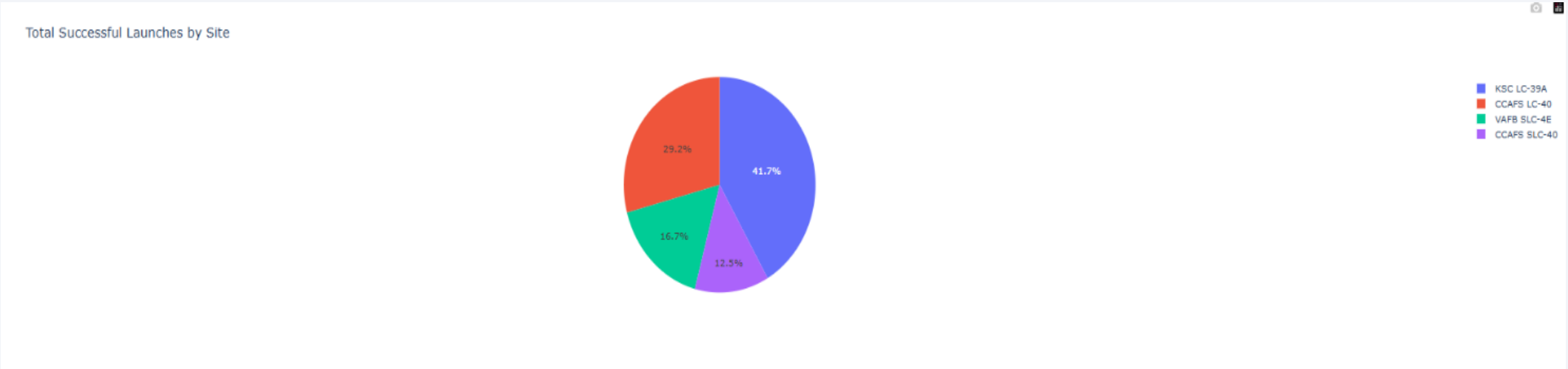
Highest successful launch site

- KSC LC-39A has the highest successful launch in all sites



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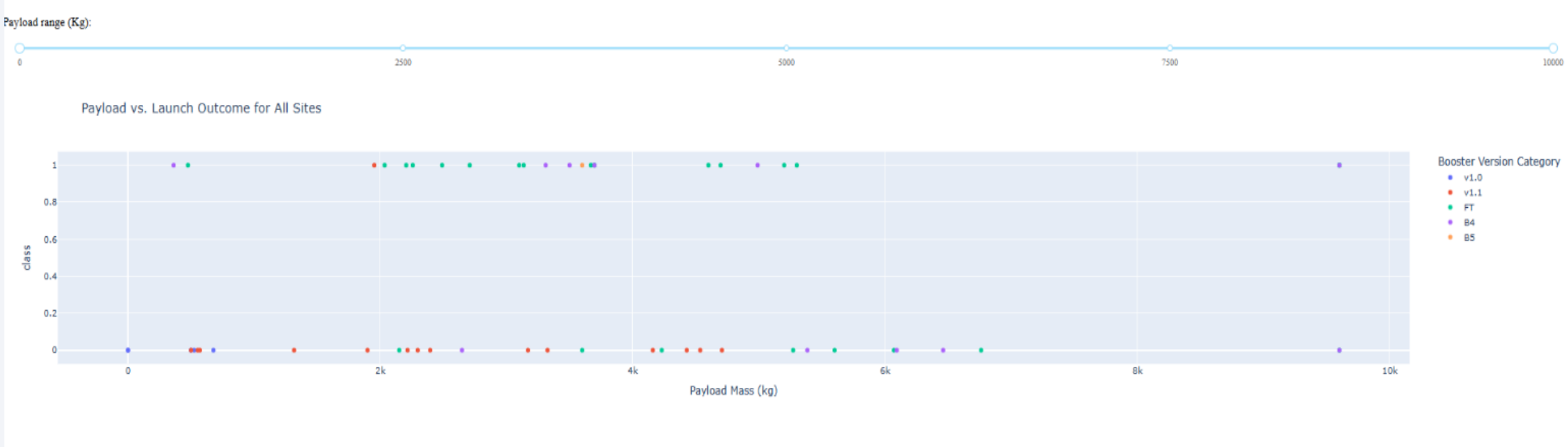
<Dashboard Screenshot 3>

- The payload range or booster version having highest success rate is FT, and the lowest is v1.1.



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Section 5

Predictive Analysis (Classification)

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Classification Accuracy

- In the classification accuracy scenario , the Logistic Regression is the best model with test accuracy rate of 0.83.
- That means the predicted value will be more accurate than any othe model.

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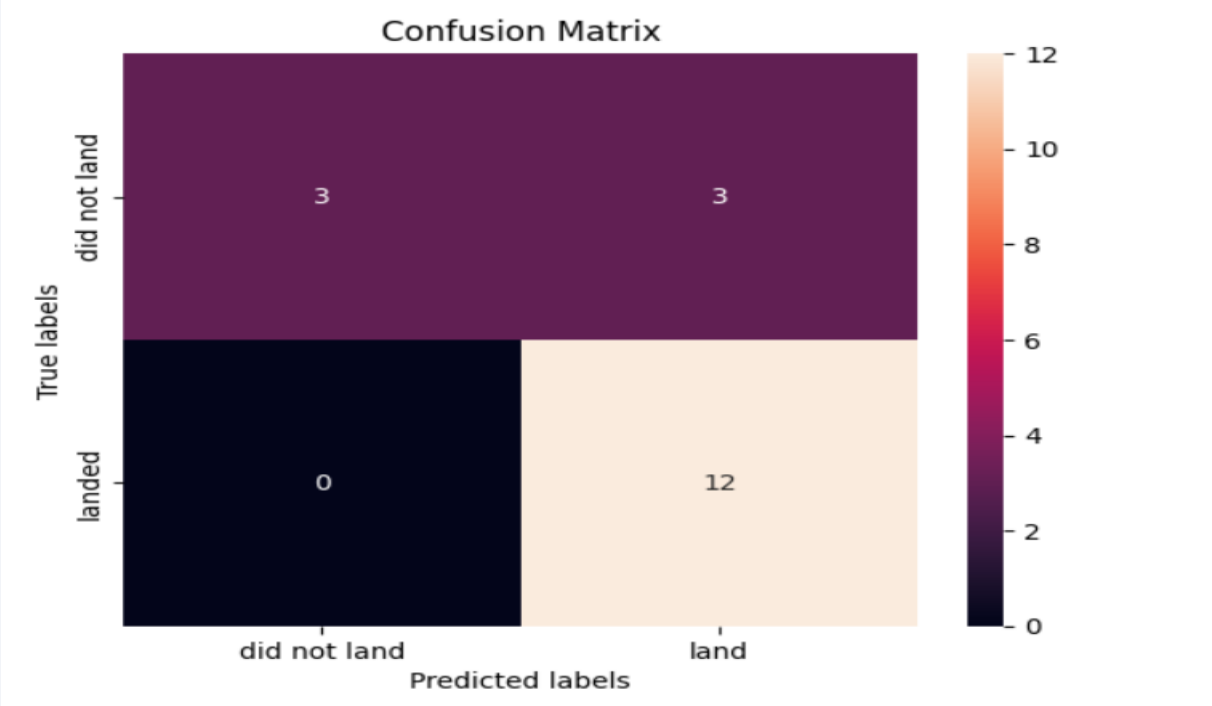
Confusion Matrix

- The best performing model is Logistic Regression model



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Conclusions

- Successfully completed the end-to-end data science workflow from **data collection and preprocessing to predictive modeling and evaluation**.
- Exploratory Data Analysis revealed key patterns influencing launch success, while interactive dashboards enabled intuitive data exploration and insights.
- Multiple classification models were built and optimized, and the **best performing model** was selected based on validation metrics and generalization performance.
- The final predictive solution demonstrates reliable accuracy and provides a **data-driven approach to forecasting mission outcomes**, supporting better decision-making.

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Appendix

- **Code Assets**
  - Python scripts for data collection and preprocessing
  - Machine learning model training and evaluation code
  - Visualization scripts for EDA charts
- **Queries & Data Processing**
  - SQL queries used for data extraction and transformation
  - Data cleaning and feature engineering steps
- **Analytical Outputs**
  - Jupyter Notebook outputs and experiment results
  - Model performance indicators and validation metrics
  - Confusion matrix and ROC curves
- **Visual Assets**
  - EDA charts and plots
  - Interactive dashboard screenshots
  - Flowcharts of model pipeline
- **Data Sources**
  - Original dataset used for analysis
  - Processed dataset for modeling
  - External references and documentation

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